

Advanced MS Access

Class 7 Computer Science Lesson Guide

 Primary Key

 Foreign Key & Relationships

 Queries

 Forms

 Reports

Quick Recap: What is a Database?

 **Database:** An organized collection of data or information stored electronically.

 **DBMS:** Software used to create, manage, and retrieve data quickly.

 **RDBMS:** Relational Database Management System (like MS Access) that stores data in multiple connected tables!



Primary Key – The VIP Unique ID!

-  **Uniquely Identifies Records:** A Primary Key is a column that uniquely identifies each row in a table (e.g., your Student Roll No. or Cust ID).
-  **No Duplicates Allowed:** No two rows can ever have the exact same Primary Key value.
-  **No Null (Blank) Values:** It can NEVER be left blank or empty!
- 1 Only One Primary Key:** Each table can have only ONE primary key field.

Foreign Key – Connecting Tables



What is a Foreign Key?

A **Foreign Key** is a field in one table that matches the **Primary Key** column of another table.

It acts like a bridge connecting two separate tables together!



Real-Life Example

In a store database:

- **Customers Table:** Cust ID is the Primary Key.
- **Orders Table:** Cust ID is used as a Foreign Key to know which customer placed which order!

How to Create Relationships

1

Open Tool

Click on **Database Tools** tab and select **Relationships**.

2

Add Tables

Select tables (e.g., Customers & Orders) in Show Table dialog & click **Add**.

3

Drag & Drop

Drag Cust ID from Customers table to Cust ID in Orders table.

4

Enforce & Create

Check **Enforce Referential Integrity** option and click **Create**!

What is a Query?



Asking Questions

A **Query** is like a question you ask the database to get specific information based on conditions filter.

Example: *"Show me all customers who live in 'Hyderabad'!"*



What is a Dynaset?

The output or result generated after running a query is displayed in tabular form, known as a **Dynaset**!

It shows only the matching records you asked for.

2 Ways to Create Queries



1. Query Wizard

A step-by-step easy wizard that guides you through selecting tables, fields, and titles effortlessly.



2. Design View

Gives complete control! You can specify exact criteria (e.g., Cust ID = 101) and filter records precisely.



Running a Query

After setting design criteria, click the red **Run (!)** button on the Design tab to view your Dynaset result!

Forms – Easy Data Entry



What is a Form?

A Form is a user-friendly window (like a display case) that shows **one record at a time**.



Why Use Forms?

It makes viewing, entering, and editing data into tables super easy, neat, and mistake-free!



Form Wizard

You can easily build forms using the **Form Wizard** under the **Create** tab step-by-step.

Reports – Print & Present Data



What is a Report? A database object used to display and print data in an organized, stylish layout.



Grouping & Sorting: You can group data (e.g., by City) and sort records in ascending/descending order.



Professional Summary: Perfect for sharing results with teachers or managers!



Quick Lesson Summary

Object / Concept	Main Function	Key Fact to Remember
Primary Key	Uniquely identifies each record in a table	Cannot be null & no duplicate values
Foreign Key	Links two tables together	Matches the primary key of another table
Query	Extracts specific data based on conditions	The result is called a Dynaset
Form	User-friendly screen to view/edit data	Displays one record at a time
Report	Organized layout for printing data	Created using Report Wizard

Class Quiz Time!

Let's check what we learned today!

Quick Review Questions:

- Q1: Can a Primary Key contain duplicate or blank values? *(Ans: No!)*
- Q2: What is the result table of a Query called? *(Ans: Dynaset!)*
- Q3: Which object is best for printing formatted summaries? *(Ans: Report!)*
- Q4: Which tab contains the Relationships button? *(Ans: Database Tools tab!)*

Image Sources



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